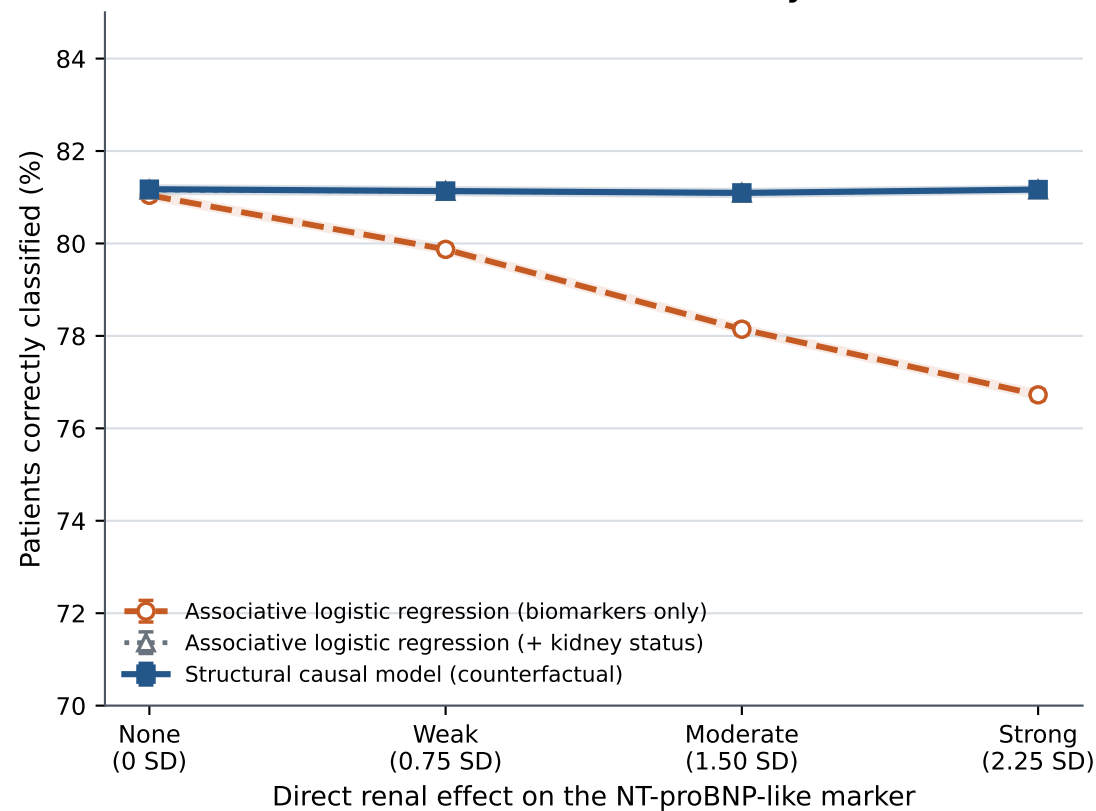
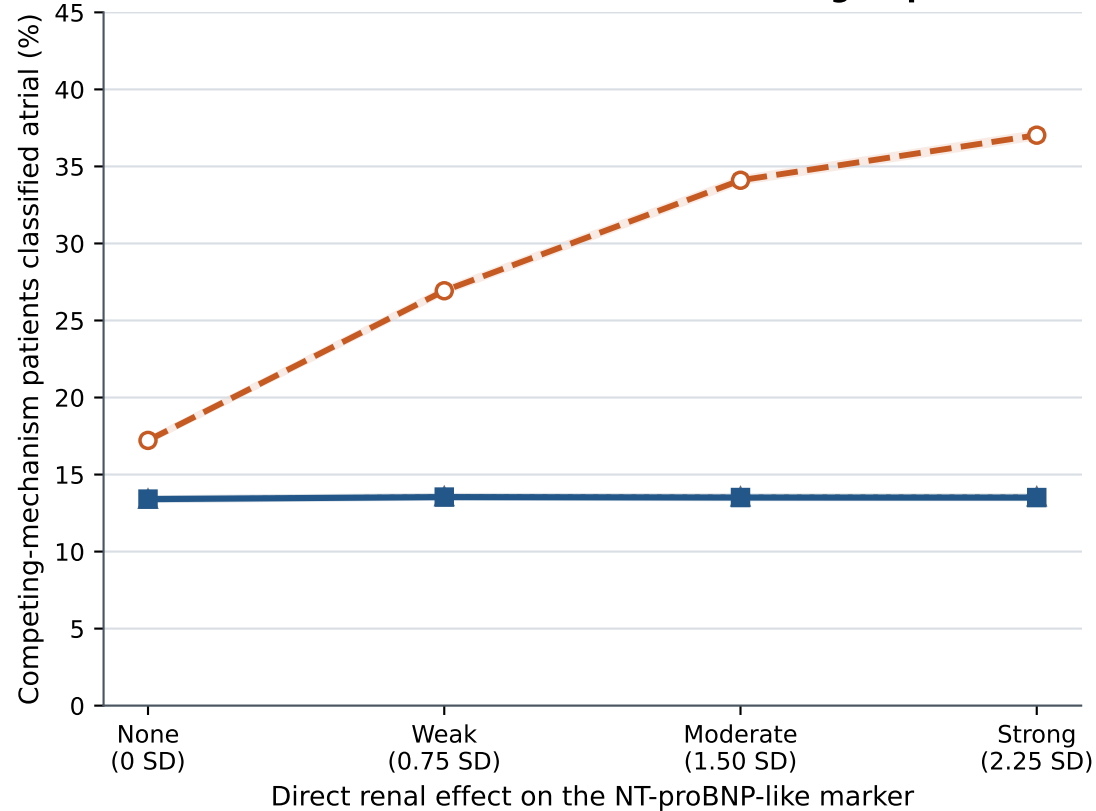


Figure P1. Associative classifiers versus a structural causal model

A. True-mechanism classification accuracy



B. False atrial classification in the renal subgroup



Means across 500 paired train/test simulations (3,000 training; 1,000 test patients per repeat); shaded bands and error bars are 95% Monte Carlo CIs. All methods use the same training and test patients.